

PHY4071 Senior Laboratory

Experiment P2: Electron Spin Resonance

Rori Bowman

Lab Partner: Niru Shakya

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Abstract

Electron spin resonance (ESR) was used to study the interaction between electron spin and a magnetic field using a DPPH sample. The resonance frequency was determined from the peak of the amplitude versus frequency curve and found to be $\nu_{\text{res}} = (33.6 \pm 1.0)$ MHz. Measurements of resonance frequency at different magnetic field strengths showed a linear relationship, consistent with the theoretical prediction that $\nu \propto B$. A linear fit to the plot yielded a slope of (24.1 ± 0.6) MHz/mT, which was used to calculate the electron g -factor. The experimental value was found to be $g_e = 1.72 \pm 0.04$, which differs from the accepted value by 14%. This discrepancy is attributed to uncertainties in the magnetic field determination. Overall, the experiment confirms the expected behavior of electron spin resonance.

1 Objectives

The objectives of this experiment are:

1. Observe electron spin resonance in a diphenyl-picryl-hydrazyl (DPPH) sample.
2. Measure the relationship between magnetic field and resonant frequency.
3. Determine the gyromagnetic ratio of spin and magnetic dipole moment (g_s) for the electron.

2 Theory

Electron Spin Resonance (ESR) is based on the interaction between a magnetic field and the intrinsic spin of the electron. An electron has spin $s = \frac{1}{2}$, which gives rise to a magnetic moment

$$\vec{\mu}_s = -g_s \mu_B \frac{\vec{s}}{\hbar}, \quad (1)$$

where g_s is the electron g -factor and μ_B is the Bohr magneton. This magnetic moment causes the electron to behave like a small magnetic dipole.

When a magnetic field $\vec{B}$ is applied, the magnetic moment interacts with the field, giving the energy

$$E = -\vec{\mu}_s \cdot \vec{B}. \quad (2)$$

Because the spin is quantized, only two orientations are allowed, corresponding to $m_s = \pm \frac{1}{2}$. This results in two distinct energy levels, a phenomenon known as Zeeman splitting. The separation between these levels is

$$\Delta E = g_s \mu_B B. \quad (3)$$

This shows that the energy difference between the spin states increases linearly with the applied magnetic field.

If an oscillating electromagnetic field of frequency ν is applied, transitions between these two spin states occur when the photon energy matches the energy difference. This gives the resonance condition

$$h\nu = g_s \mu_B B. \quad (4)$$

Energy is absorbed by the system, producing a detectable signal. This resonance condition forms the basis of ESR.

The magnetic field is produced by a pair of Helmholtz coils. The field at the center of the coils is given by

$$B = \mu_0 \left(\frac{4}{5} \right)^{3/2} \frac{nI}{r}, \quad (5)$$

where n is the number of turns, I is the current, and r is the radius of the coils. This allows the magnetic field to be controlled through the current. Combining these results shows that the resonance frequency is proportional to the magnetic field,

$$\nu = \frac{g_s \mu_B}{h} B. \quad (6)$$

Therefore, a plot of ν versus B should be linear, and the slope of this relationship provides a direct way to determine the electron g -factor:

$$g_s = \frac{h}{\mu_B} \left(\frac{\nu}{B} \right). \quad (7)$$

Thus, by measuring how the resonance frequency changes with magnetic field, the magnetic properties of the electron can be experimentally determined.

3 Equipment

3.1 Apparatus List

The principal equipment used in this experiment included:

- ESR base unit
- ESR adapter
- Helmholtz coils
- High-frequency oscillator
- DPPH sample
- Oscilloscope
- Digital multimeter
- DC and AC power supplies

4 Procedures

1. Set up the ESR apparatus with the Helmholtz coils and place the DPPH sample at the center within a coil.
2. Connect the oscillator and oscilloscope to the ESR unit.
3. Vary the frequency and record the signal amplitude to obtain the resonance curve.
4. Determine the resonance frequency from the peak of the curve.
5. Adjust the current through the coil at different frequencies until resonance is observed.
6. Record the voltage, calculate the current, and determine the magnetic field.
7. Plot ν vs B and use the slope to calculate the electron g -factor.

5 Data, Analysis, and Results

5.1 Experimental Data

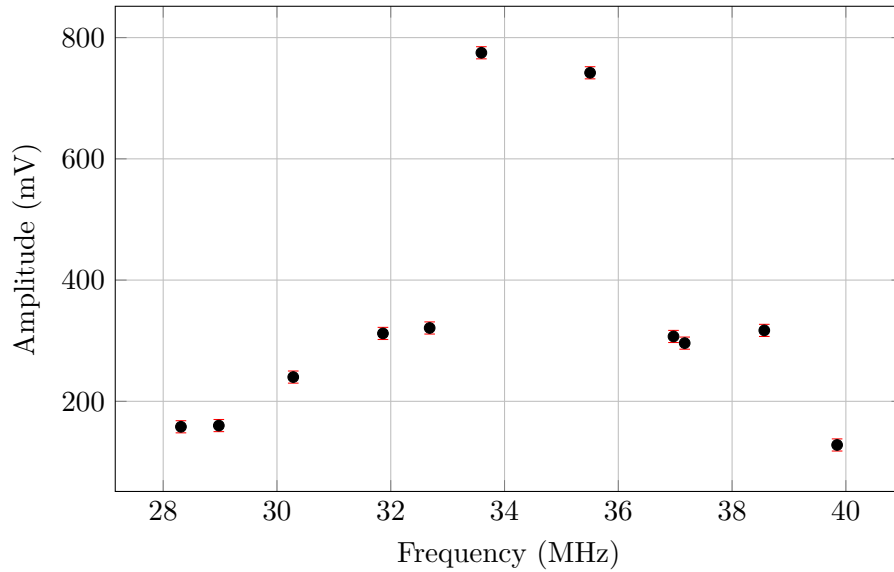


Figure 1: Resonance curve showing amplitude as a function of frequency.

The highest peak of the resonance curve is where the resonance frequency is located. Using the data from Figure 1, the resonance frequency can be identified around the peak $\nu \approx 33.6$ MHz. We must account for error when performing data analysis.

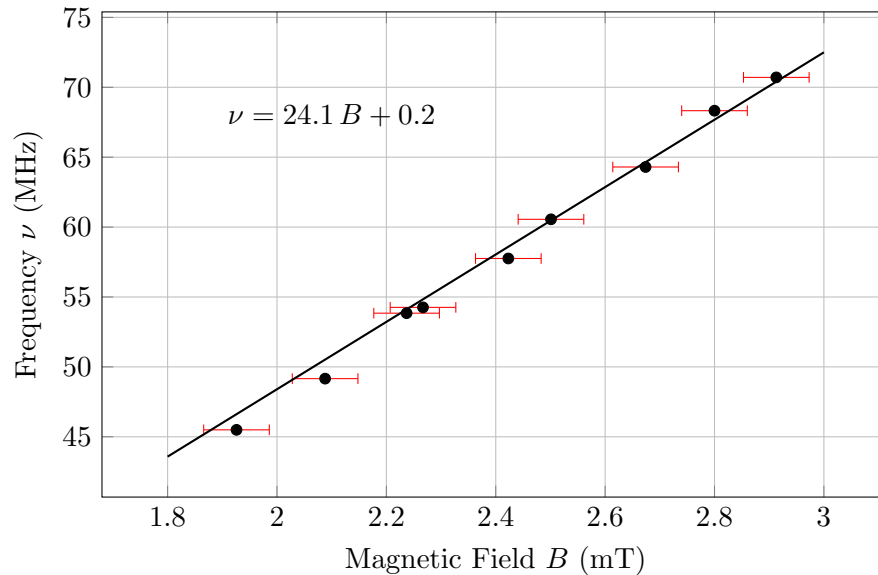


Figure 2: Resonance frequency as a function of magnetic field with a linear fit.

The data points closely follow the linear fit, indicating good agreement with the expected linear relationship. The slope of Figure 2 is how we obtain the g -factor. Also note that the intercept of

the fit is close to zero, as expected from the theoretical relationship $\nu \propto B$. This indicates that there is no significant offset or error in the measurement.

5.2 Results, Analysis, and Comparison with Theory

Resonance Frequency with Uncertainty

The resonance frequency was determined from the peak of the amplitude versus frequency curve in Figure 1. Since the peak is not sharp, the uncertainty was estimated from the width of the top region of the curve. The two highest points near the peak occur at

$$\nu_1 = 33.593 \text{ MHz}, \quad \nu_2 = 35.504 \text{ MHz}.$$

The uncertainty in the resonance frequency was taken as half the difference between these values, which gives

$$\Delta\nu = \frac{\nu_2 - \nu_1}{2} = \frac{35.504 - 33.593}{2} = 0.956 \text{ MHz}.$$

Therefore,

$$\Delta\nu \approx \pm 1.0 \text{ MHz}.$$

The uncertainty from the frequency display is

$$\Delta\nu_{display} = \pm 0.001 \text{ MHz},$$

which is significantly smaller and negligible compared to the uncertainty from the peak width. So, the resonance frequency accounting for error is

$$\nu_{\text{res}} = (33.6 \pm 1.0) \text{ MHz}$$

g-factor with Uncertainty

Figure 2 shows the resonance frequency as a function of the applied magnetic field. The data exhibit a clear linear relationship, consistent with the expected relationship

$$\nu = \frac{g_s \mu_B}{h} B.$$

A linear fit to the data gives

$$\nu = (24.1 \text{ MHz/mT}) B + 0.2.$$

The slope of this fit represents the proportionality between frequency and magnetic field:

$$m = \frac{g_s \mu_B}{h}.$$

So, the electron g -factor can be determined from the slope using

$$g_s = \frac{h}{\mu_B} m.$$

Plugging in values for $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$ and $\mu_B = 9.274 \times 10^{-24} \text{ J/T}$,

$$m = 24.1 \frac{\text{MHz}}{\text{mT}} = 24.1 \times 10^9 \frac{\text{Hz}}{\text{T}},$$

gives

$$g_s = \frac{6.626 \times 10^{-34}}{9.274 \times 10^{-24}} (24.1 \times 10^9).$$

Evaluating,

$$g_s \approx 1.72.$$

The dominant uncertainty arises from the magnetic field measurement. Since the magnetic field is proportional to the current, and the current is determined from the measured voltage,

$$\frac{\Delta B}{B} = \frac{\Delta V}{V}.$$

Using $V \approx 0.40$ V with uncertainty $\Delta V = 0.01$ V,

$$\frac{\Delta B}{B} \approx \frac{0.01}{0.40} = 0.025.$$

So, the fractional uncertainty in the slope is approximately 2.5%, which gives

$$\Delta m \approx 0.025(24.1) \approx 0.6 \text{ MHz/mT}.$$

Since $g_s \propto m$, the same fractional uncertainty applies:

$$\frac{\Delta g_s}{g_s} = \frac{\Delta m}{m}.$$

Therefore,

$$\Delta g_s \approx 0.025(1.72) \approx 0.04.$$

The final result is

$$\boxed{g_s = 1.72 \pm 0.04}$$

Comparison with Theory

The accepted value of the electron g -factor is $g_e \approx 2.00$. The experimental result is lower than expected, most likely due to uncertainties in the magnetic field calibration and measurement of the current. However, the relationship between frequency and magnetic field is clearly observed to be linear, therefore confirming the expected behavior of electron spin resonance.

The percent error between the experimental value and the accepted value of the electron g -factor is given by

$$\% \text{ error} = \frac{|g_{\text{exp}} - g_{\text{theo}}|}{g_{\text{theo}}} \times 100.$$

Using $g_{\text{exp}} = 1.72$ and $g_{\text{theo}} = 2.00$:

$$\% \text{ error} = \frac{|1.72 - 2.00|}{2.00} \times 100.$$

$$\% \text{ error} = \frac{0.28}{2.00} \times 100 \approx 14\%.$$

Final Results

Quantity	Value
Resonance Frequency ν_{res}	(33.6 ± 1.0) MHz
Slope $(\frac{\nu}{B})$	(24.1 ± 0.6) MHz/mT
Exp. g -factor g_{exp}	1.72 ± 0.04
Theo. g -factor g_{theo}	2.00
g -factor % Error	14%

Table 1: Experimental results and comparison with the theoretical value.

6 Conclusions

In this experiment, electron spin resonance was successfully observed using a DPPH sample. The resonance frequency was determined from the peak of the amplitude versus frequency curve to be $\nu_{res} = (33.6 \pm 1.0)$ MHz. A linear relationship between resonance frequency and magnetic field was observed, consistent with the theoretical prediction $\nu \propto B$. The slope of the linear fit was used to calculate the electron g -factor, yielding $g_e = 1.72 \pm 0.04$.

This value differs from the accepted value of $g_s \approx 2.00$ by approximately 14%, indicating that the result does not agree within experimental uncertainty. The discrepancy is due to systematic errors, likely in the determination of the magnetic field from the coil current. Additional sources of uncertainties include limitations in measuring the current and identifying the exact resonance frequency from the wide peak.

Despite this deviation, the experiment clearly demonstrates the expected linear dependence of resonance frequency on magnetic field, confirming the underlying physics of electron spin resonance. Overall, the results support the theoretical model while highlighting the sensitivity of the g -factor measurement to experimental uncertainties.

A Raw Data

$f/1000$	Frequency (Hz)	Amplitude (mV)
28312	28312000 ± 1000	158 ± 10
28978	28978000 ± 1000	160 ± 10
30286	30286000 ± 1000	240 ± 10
31863	31863000 ± 1000	312 ± 10
32684	32684000 ± 1000	321 ± 10
33593	33593000 ± 1000	775 ± 10
35504	35504000 ± 1000	742 ± 10
36970	36970000 ± 1000	307 ± 10
37167	37167000 ± 1000	296 ± 10
38566	38566000 ± 1000	317 ± 10
39846	39846000 ± 1000	128 ± 10

Table 2: Raw data used for the resonance curve.

$f/1000$	ν (MHz)	V_{rms} (V)	I (A)	B (mT)
54257	54.257 ± 0.001	0.379 ± 0.01	0.536	2.267 ± 0.06
64298	64.298 ± 0.001	0.447 ± 0.01	0.632	2.674 ± 0.06
49159	49.159 ± 0.001	0.349 ± 0.01	0.494	2.088 ± 0.06
45497	45.497 ± 0.001	0.322 ± 0.01	0.455	1.926 ± 0.06
70707	70.707 ± 0.001	0.487 ± 0.01	0.689	2.913 ± 0.06
53842	53.842 ± 0.001	0.374 ± 0.01	0.529	2.237 ± 0.06
57754	57.754 ± 0.001	0.405 ± 0.01	0.573	2.423 ± 0.06
68324	68.324 ± 0.001	0.468 ± 0.01	0.662	2.800 ± 0.06
60556	60.556 ± 0.001	0.418 ± 0.01	0.591	2.501 ± 0.06

Table 3: Raw data used for the plot of resonance frequency versus magnetic field.